

Cross-Project Bug Localization Paper: Step-by-Step Actions

To A Eashaan Rao. From Sridhar Chimalakonda. RISHA Lab, IIT Tirupati. Reference: version 7 of the manuscript. Target venue: Journal of Systems and Software.

Core Argument

Version 7 is a clear improvement and the framing work has largely landed. Six steps remain. Step 1 is the only one that can change numbers, so it comes first and everything else waits on it. Step 2 is a single ablation that needs no training and protects our most useful finding. Steps 3 to 6 are rewriting, and all of them make the paper shorter rather than longer. The paper is now 30 pages, and length is itself a risk: it enlarges the surface a reviewer can object to and it multiplies the places where the paper can disagree with itself. Target 24 pages.

Step 1. Reconcile the metrics with the evaluation code

This is the one blocking item and it has been open across two versions. It is a code-reading task, not an experiment. Please do it before touching anything else, because if the definitions change, parts of Section 4 change with them.

The problem, stated two ways

First, the Top-K bound. For a bug report with a single relevant file, MRR cannot exceed $\text{Top1} + (\text{Top5} - \text{Top1})/2 + (\text{Top10} - \text{Top5})/6 + (1 - \text{Top10})/11$. Applying this to Table 6, TRANP-CNN reports MRR 0.326 against a maximum of 0.262 for WP-small, 0.389 against 0.291 for WP-large, and 0.380 against 0.288 for CP-transfer. BLAZE and COOBA satisfy the bound in every row.

Second, the retrieval ceiling. Table 3 now reports Recall@300 and its caption states that this bounds every absolute metric for that project. jupyterlab is 47.0 percent. In Case 2, on a jupyterlab target, TRANP-CNN reports 0.654, 0.760 and 0.800, and BLAZE reports 0.512. All four exceed the ceiling.

Every violation sits on a jupyterlab target, and jupyterlab is the target in twelve of the sixty-three pairs. That pattern points to a definitional mismatch rather than a training problem, which is the likely good news here.

What to check, in the evaluation script

1. Is MRR averaged over all test bugs, or only over the bugs whose ground truth appeared in the FAISS top 300? If the latter, that single convention explains both symptoms at once.
2. Is Top-K defined as at least one ground-truth file in the top K, or all ground-truth files in the top K? Section 3.7 now says most bug reports have more than one ground-truth file, so this matters and is currently unstated.
3. Is Recall@300 computed over the full bug corpus or over the 20 percent test split, and under which of the two multi-file conventions above?

Cheapest single diagnostic: verify jupyterlab's Recall@300 of 47.0 percent first. If that one figure is wrong, most of the problem disappears.

What to do with the answer

If MRR and MAP are conditioned on retrieval while Top-K and Recall are not, the numbers are fine and the fix is two sentences in Section 3.7 stating the convention, plus a note that MRR and MAP are recall-conditioned and therefore not comparable across projects with differing retrieval recall. Nothing else moves.

If it is anything else, stop and tell me before continuing. That case may move numbers in Table 6, Table 8 and all three illustrative cases, and the rest of this note would need rethinking.

Either way, Section 3.7 must describe what the code computes, not what is correct in the abstract. Version 7 moved these definitions toward textbook correctness, which widened the gap rather than closing it. Also carry the multi-file ground truth through Section 3.4, which still describes a singular `gt_file_path`, and through the

Top-K and Recall@300 definitions.

Step 2. Run one ablation: untrained BGE with MaxSim

No training. The BGE embeddings exist, the FAISS indexes exist, and BLAZE's MaxSim scoring code exists. This is re-scoring the same candidate pool with the adapters removed. Hours of GPU time across the thirteen targets, and it can run while you work on Steps 3 onward.

Why it matters. Retrieval pools candidates using mean-pooled BGE embeddings, and BLAZE then reranks that same pool with MaxSim over the same underlying signal. So the cold-start comparison against FAISS at 0.029 confounds two changes: source-side adapter training, and the switch from mean pooling to MaxSim. Section 5.4 already identifies this and calls the missing baseline the most valuable addition for follow-up. A reviewer who reads that will ask why we named the decisive check and did not run it when it needs no training.

If untrained MaxSim lands well below 0.262, the cold-start finding becomes considerably stronger than it is now and the awkward paragraph in 5.4 goes away. If it lands near 0.262, we need to know that before a reviewer does, and the claim gets rewritten as a retrieval-engineering result, which is still publishable and still useful.

BM25 is worth adding if it fits alongside, since Section 2.1 identifies IR methods as requiring no training data and therefore as the real zero-label alternative. It is CPU-only and roughly a day. If it does not fit, skip it and add one sentence to 5.4 stating that classical IR was not evaluated as a zero-label comparator, so the cold-start result is relative to WP-small rather than to the full set of options open to an unlabelled project.

Do not attempt the COOBA original-benchmark run or chronological splits now. Both are larger, neither bears on a headline claim, and the fidelity paragraph you already wrote in 5.4 covers the first well.

Step 3. Cut the paper to about 24 pages

Do this before the claim sweep, because several of the cuts delete overclaims as a side effect and there is no point editing text that is about to go.

1. Delete the three RQ summary boxes. They restate numbers from the tables directly above them, then Section 4.4 restates those, then the conclusion restates those again. Four statements of the same findings. Keep 4.4. Saves about two pages.
2. Cut the introduction summary box to the contributions list plus the key insight sentence. The What works, What doesn't and Why blocks duplicate the abstract. Saves most of a page.
3. Section 4.5, keep Case 2 only. It is the only one showing something the aggregates cannot. Case 1 is the cold-start finding on one pair and Case 3 is the WPL exceedance on one pair, both already covered by Table 8 and the TOST results. This also removes the Case 3 table, whose values sit above the retrieval ceiling. Saves about a page and a half.
4. Section 5.1, six recommendations down to four. Items 1, 3, 5 and 6 are distinct. Item 2 restates Table 8 and item 4 restates the TOST result.
5. Section 5.2, cut the two-competing-hypotheses passage to a short paragraph. It argues at length against a confound that 5.4 then concedes is not controlled, and the length weakens rather than strengthens it.
6. Section 2, tighten. The agentic-LLM paragraph in 2.2 is a page of scoping for a paradigm we do not evaluate; three sentences will do. 2.1 and 2.3 both run long on material that does not bear on the design. Saves about a page.
7. Section 3.6, let Table 5 carry WP-large and cold-start. Keep the prose bullets only for WP-small and CP-transfer, where the design rationale matters.
8. Merge the duplicated methodological-principle statement. Section 3's opening and the paragraph beginning "As argued in Section 1" say the same thing on one page.

Step 4. Sweep the remaining overclaims

About eighteen sentences across fourteen locations. Every one is a deletion or a single-word substitution. Nothing needs rechecking and every fix shortens the paper. Rather than working from a list, search the source for these five strings and inspect every hit.

confirm establish robust no existing / every published / not answered anywhere
removing the need

That catches most of them. The word confirm is now the dominant pattern, with six instances, four of them in figure captions, which is why the last sweep missed them. Reviewers read captions. Two examples: the Figure 7 caption reads a non-significant correlation as confirming architectural independence, and 4.2.4 reads 17 of 28, about p equal to 0.17, as confirming a LoC-driven failure mode.

Six named items the search will not catch:

1. Section 4.3.1 body. It still carries the full strong form, including that CPL benefit is not a property of the pair alone, two paragraphs after reporting ρ equal to 0.559 at p below 0.001. Rewrite to the agreed sentence, as already done in the abstract and elsewhere. Section 4.4's fifth bullet needs the same.
2. Independently replicated, in 4.1.2 and the RQ1 summary. Two models agreeing is not replication, particularly when they are the two that share a ranking family.
3. Any within-domain source project, in 4.1.4 and the conclusion, contradicted by our own 74.6 percent and by the two negative-transfer cases.
4. The italicised sentence that within-project source performance is actively misleading, in 4.3.2 and repeated in 5.1. Remove the italics and the second sentence. It rests on p equal to 0.019 uncorrected, one model, and a re-implementation.
5. Indicative upper bound in 4.2.1. Group means are not bounds. Use typical gain observed in this corpus.
6. Allows conclusions about why CPL succeeds or fails, in Sections 1 and 3. The design supports attributing outcome differences to the models. It does not establish mechanism.

Three more worth naming: a medium effect means the advantage is real and reproducible in 4.1.2, against a single seed and a reported spread reaching 0.29 MRR; essentially zero adoption risk in 4.2.2, from two pairs with a median loss below the noise floor; and in Case 2, that a practitioner using BLAZE would correctly conclude ansible is the optimal source, where both correctly and optimal overreach from a single pair whose WP-small run the paper itself flags as an outlier.

Also finish the paradigm-to-model substitution begun in 3.2. Still outstanding: 2 of 3 benchmarked architectural paradigms in 4.1.2 and the RQ1 summary, the sentence beginning We classify an architectural paradigm, and most of 4.2.1 and 4.2.3.

Step 5. Fix the contradictions introduced in version 7

Three of these came from text added to satisfy earlier comments, which is why they need naming explicitly.

1. Joint versus sequential training. Section 3.6 now states plainly that CP-transfer is jointly trained in the same step and not a pretrain-then-fine-tune schedule. Four passages still describe initialisation and pre-training: 4.2.1, 5.2 twice, Case 3, and the conclusion. Keep 3.6's version, since it describes what the code does, and rewrite the mechanism argument in 5.2 in joint-training terms. This matters because the paper's Why rests on that argument.
2. Figure 3's caption asserts that every cell exceeds the 0.20 viability threshold. Two cells read 0.187 and 0.170.
3. Table 7's tie footnote gives 27 wins for COOBA Top-1, which is 42.9 percent against the table's 43.5 percent, and 30 wins for TRANSP-CNN Top-10, which is 47.6 percent against the table's 46.0 percent. The footnote was added to buy credibility on exactly this point.

4. Table 9's jupyterlab exclusion. Only cross-domain figures are given, but Group A is DS by DS and jupyterlab is a DS project, so within-domain changes too. And the caption says TRANP-CNN's domain effect weakens further, when 76.7 against 36.8 is a 39.9 point gap, wider than the 28 points in the table. The exclusion strengthens TRANP-CNN's domain sensitivity. Please recompute both columns and correct the reading.
5. Three broken sentences: the missing H symbol in Section 3.8 rendering as the -divergence theory; the sentence in 5.4 beginning our smallest WP-small training set is themselves a lower bound, where the logic is also muddled since a training set size does not bound a holdout size; and in 4.2.4, For The CNN-based reranker (TRANP-CNN) has a consistency rate.

Step 6. Close out the venue requirements

About an hour in total, and these are the only items on the whole list that can cost us before a reviewer opens the paper.

1. Highlights, three to five, each within 85 characters.
2. CRediT author contribution statement.
3. Declaration of competing interest.
4. Formal funding statement.
5. RISHA Lab AI assistance disclosure.
6. Complete the ACM SIGSOFT empirical standards checklist for the General and Data Science standards, cite Ralph and colleagues 2021, and include the completed checklist in the Zenodo package. Add a short paragraph in Section 3 naming which standards the study follows.
7. Name the COOBA and TRANP-CNN implementations explicitly in Data Availability. Given the re-implementation threat they are the most important artifact in the package.

Two small analytical checks, if time permits. Both use only the existing results file. First, correlate target label count against CP-transfer MRR alone, not just against gain, since gain is CPT minus WPS and WPS is trained on 20 percent of that same label count, so part of rho equal to minus 0.77 may be regression on the subtrahend. If it survives, the RQ2 finding is much stronger; if not, one sentence acknowledging the coupling is enough. Second, report the residual variance and the ICC alongside the group variance of 0.003 in the mixed-effects paragraph, since small relative to the residual cannot currently be checked.

What we are deliberately not doing

Add a short closing paragraph to Section 5.4, three or four sentences, naming the untrained baseline if we do not run it, the IR comparator, the COOBA original-benchmark run and the chronological split, with one line each on which conclusion it would sharpen. Reviewers respond well to that when the claims have already been scoped to survive without those checks, and badly when they have not. That paragraph is also our response-letter skeleton if a reviewer asks for any of the four.

Sequence and Timing

Week	What
1	Step 1 metric check, and start the Step 2 ablation running in parallel
2	Step 3 cut pass, then Step 4 claim sweep
3	Step 5 contradictions, Step 6 venue items, send to Ch
4	Ch review pass and submit

Two rules for this round. Do not begin Step 3 until Step 1 is settled, or you will soften claims around figures that then change and pay for it twice. And this is the last full round: every version so far has been additive, and version 7 closed six contradictions while opening four, three of them in text written to satisfy review comments. Steps 3 to 5 are subtractive by design, which is how that pattern stops.

The study itself is sound and the protocol is a real contribution. What we are closing is the distance between what we measured and what the paper says we showed. Please come back to me on Step 1 this week before anything else.

Sridhar Chimalakonda

The original ideas, analysis, and research synthesis in this report are the intellectual contribution of the authors. AI-assisted tools were used to enhance the clarity and presentation of the writing.